Script 02

- Instantiating primitive models.
- Illumination and shadows.
- Animation.
- Perspective camera vs orthographic camera
- Window resizing
- Controlling the camera

1.1 Instantiating primitive models

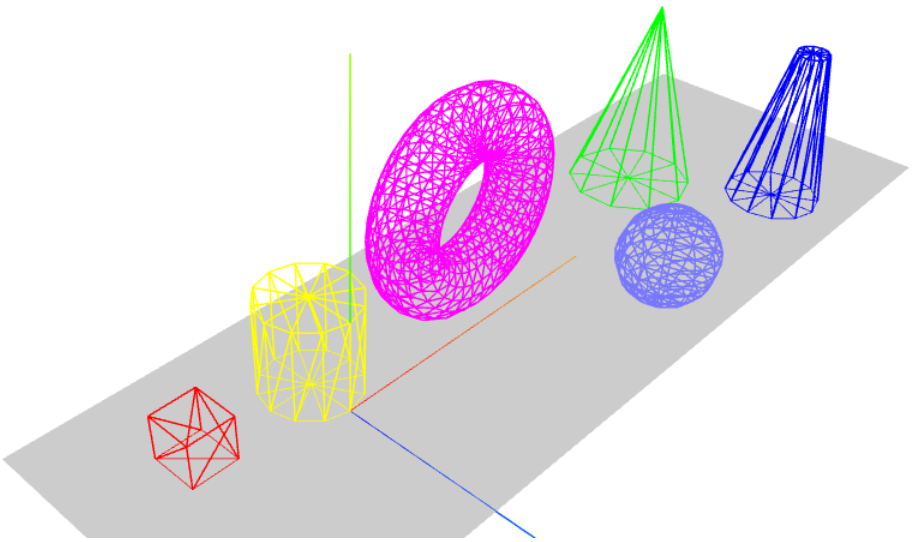
Open the file **threejs_ex_02_01.html**

Analyze the **init()** function:

- How many **models** are defined in the scene?
- Where are they placed? Are any **rotations** applied? Why?
- How is the **camera** looking at the scene?

Tasks:

- Check the **code comments** and carry out the **suggested tasks**.
- Add a **GridHelper** to the ground plane, which will help you correctly place models on the scene.
- Add **four models** to the scene – with appropriate features – to obtain the scene displayed below.



1.2 Illumination and Shadows

Simple illumination effects are easily obtained by **adding a spotlight** to the scene, assigning proper **materials** to the models, and **enabling the rendering of shadows**.

Tasks:

- Disable the **wireframe** rendering mode. Add a **spotlight** to the scene, placed **at (-40, 60, -10)**. Do you notice any **differences**? Why?

- Change the material defining the models to **Lambert Material**, which computes shading using the Gouraud model. Which **differences** do you notice?

- Add an **Ambient Light** component to scene. Disable and enable the spotlight. Which **differences** do you notice?

- Enable the rendering of shadows:

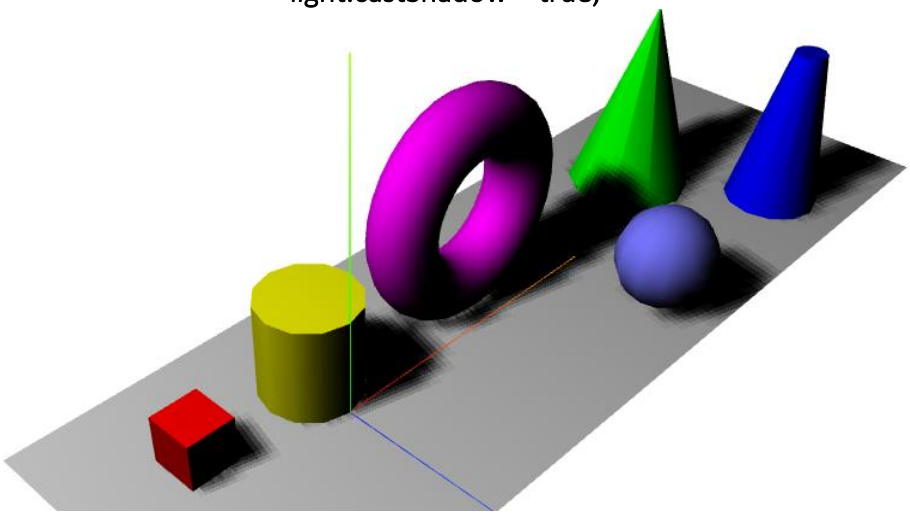
```
renderer.shadowMap.enabled = true;
```

- Enable the models to cast shadows on each other and to receive shadows:

```
plane.receiveShadow = true;
cone.castShadow = true;
cone.receiveShadow = true;
```

- Enable the light source to cast shadows:

```
light.castShadow = true;
```



- Change the material defining the models to **Phong Material**, which computes shading using the Phong model and simulates the reflection of shiny surfaces. Associate **shinier materials** to some of the models.

- Compare side-by-side the results of the Gouraud and the Phong shading models. Do you notice **any differences**?

1.3 Animation

Simple animation effects are easily obtained by **updating model position or rotation angles** just **before rendering each frame**, and by rendering an appropriate number of frames per second.

Tasks:

- Add the following code at the end of the init() function:

```
var step = 0;
// Update model features and render the scene
renderScene()

function renderScene() {
	// Rotate the cube around its axes
	cube.rotation.x += 0.02;
	cube.rotation.y += 0.02;
	cube.rotation.z += 0.02;

	step += 0.04;
	// Bounce the sphere up and down
	sphere.position.x = 20 + (10 * Math.cos(step));
	sphere.position.y = 3 + (10 * Math.abs(Math.sin(step)));

	// Render using requestAnimationFrame
	requestAnimationFrame(renderScene);
	renderer.render(scene, camera);
}
```

- What happens?

- Add code to **rotate the torus** around its XX axis and to **shuffle the cylinder** back-and-forth in the ZZ direction.

- Add code to **displace the camera** back-and-forth along a given direction.

1.4 Perspective camera vs orthographic camera

There are two different camera types in Three.js: the orthographic camera (at an indefinite distance to the scene) and the perspective camera (at a finite distance to the scene), which produce different final images.

Tasks:

- Check the documentation on the Orthographic Camera. How do you **set up** an Orthographic Camera? What is the meaning of the camera **constructor arguments**? How do you define the **viewing direction**?

- Change the camera to the Orthographic Camera. Which **differences** do you notice in a **still image** and in an **animation**?

- Compare side-by-side the images produced by the two cameras. Which camera produces **more realistic** images?

1.5 Adding an event listener to handle browser window resizing

Updating the display whenever the browser window is resized can be easily done by registering the corresponding event-handling function:

```
window.addEventListener('resize', onResize, false);
```

In this **onResize()** function, camera aspect ratio and renderer window size are updated as follows:

```
function onResize() {
        camera.aspect = window.innerWidth / window.innerHeight;
        camera.updateProjectionMatrix();
        renderer.setSize(window.innerWidth, window.innerHeight);
}
```

Tasks:

- Add the event-handling code to your example file.

- Declare the camera and renderer variables as **global variables**.

- Resize the browser window and see what happens.

- Run the example on your **smartphone**!

1.6 Camera control – OrbitControls

The **three.js library** has a large set of different **addons** which expand its basic functionality. To enable importing addons, change your code as follows:

```
"imports": {
        "three": "https://threejs.org/build/three.module.js",
        "three/addons/": "https://threejs.org/examples/jsm/"
}
```

Orbit Controls allow the camera to orbit around a target, for instance, to allow detailed model inspection. The camera is controlled with the **left mouse button** (orbiting), the **mouse wheel** (zoom) and the **right mouse button** or arrow keys (pan). On a **touchscreen**, the usual finger interaction can be used. Try it on your **smartphone**.

To use OrbitControls, import it as follows:

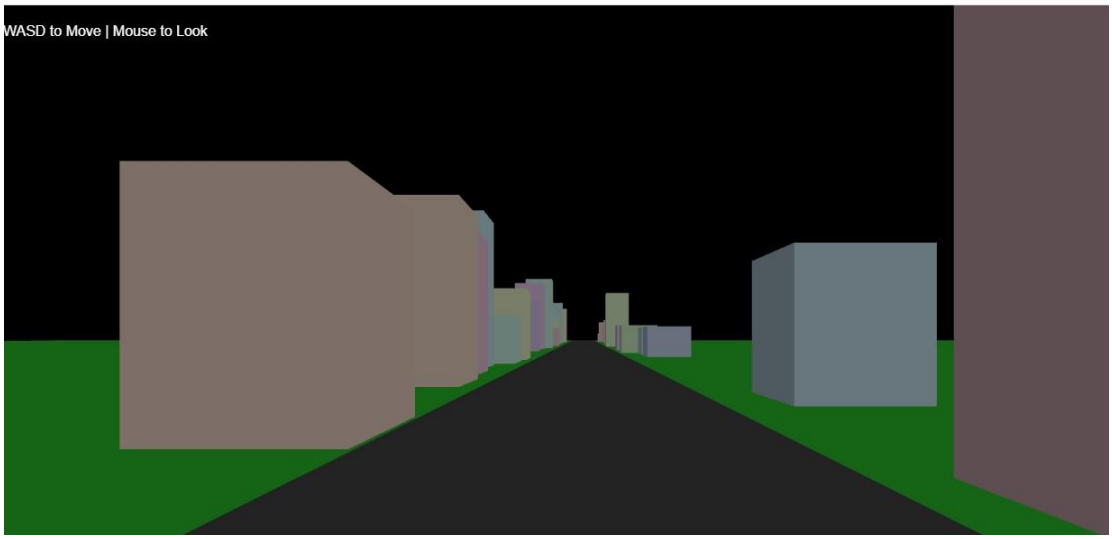
```
import * as THREE from 'three';
import { OrbitControls } from 'three/addons/controls/OrbitControls.js';
```

To control the camera, change your code as follows:

```
        // --- INTERACTION ---
        const controls = new OrbitControls(camera, renderer.domElement);
        controls.enableDamping = true;
…
        function animate() {
                requestAnimationFrame(animate);
                controls.update();
                renderer.render(scene, camera);
        }
```

1.7 New example: street walkthrough

Develop a new example, that allows you to walk through a street with simple buildings of different colors. Create a scenario similar to the one the figure.



1.8 Camera control – FirstPersonControls

First Person Controls allows the camera to move in the scene in first person walkthrough mode and to look around. It is possible to restrict the allowed camera movements.

To use FirstPersonControls, import it as follows:

```
import * as THREE from 'three';
import { FirstPersonControls } from 'three/addons/controls/FirstPersonControls.js';
```

To control the camera, change your code as follows:

```
scene = new THREE.Scene();
clock = new THREE.Timer(); // To use First Person Controls
clock.connect(document); // use Page Visibility API
…
// --- FIRST PERSON CONTROLS to simulate walking---
controls = new FirstPersonControls(camera, renderer.domElement);
controls.movementSpeed = 30;
controls.lookSpeed = 0.1;
controls.lookVertical = false; // Keep view horizontal like walking
…
```

```
function animate() {
	requestAnimationFrame(animate);
	clock.update();
	const delta = clock.getDelta();
	controls.update(delta);
	renderer.render(scene, camera);
}
```